# Prodigy Anesthesia Quick Review

## Digoxin

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| What is digitalis? | It is the term used for the genus name of the family of plants in which cardiac glycosides occur naturally.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 215.* |
| Describe the impact digoxin has on resting membrane potentials of cardiac cells. | Digoxin decreases the resting membrane potentials of cardiac cells through the inhibition of the sodium pump. The sodium pump is vital in the maintenance of normal gradients of potassium and sodium ions. This inhibition decreases the resting membrane potential, which in turn increases the excitability (automaticity) of the cardiac cells by altering the potassium ion gradient.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 312.* |
| What is the prototypical cardiac glycoside? | Digoxin<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 216.* |
| How does digoxin affect cardiac contractility? | Digoxin increases cardiac contractility.<br>*Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 214.* |
| How does digoxin increase cardiac contractility? | Through the increase of intracellular concentrations of calcium ions in cardiac cells. Inhibition of the sodium pump increases the concentration of intracellular sodium. This increased concentration of sodium causes a reduction in the extrusion of calcium ions from the sodium pump.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 216.* |
| How does digoxin affect the sodium-potassium adenosine triphosphatase ion transport system (also known as the sodium pump)? | It reversibly and selectively inhibits the sodium pump.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 312.* |
| How does digoxin affect the conduction velocity through the AV node at therapeutic dosages? | It reduces it.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 217.* |
| Is digoxin metabolized to a great extent in humans? | No. A large amount of digoxin (almost two thirds) is excreted by the kidneys unchanged.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 216.* |

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| Plasma clearance of digoxin is primarily dependent upon what organ? | The kidneys<br><br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 313.* |
| How does creatinine clearance affect renal clearance of digoxin? | Renal clearance of digoxin is proportional to creatinine clearance. Therefore, the elimination half-time is reduced in proportion to the reduction in creatinine clearance in the presence of renal dysfunction.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 214.*<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 216.* |
| Does digoxin affect the autonomic nervous system? Are these effects more predominant in the sympathetic or parasympathetic nervous system? | Yes, digoxin affects the autonomic nervous system. These effects are more prevalent in the parasympathetic nervous system. Effects are manifested as a reduction in heart rate caused by decreased activity of the SA node and prolongation of the effective refractory period.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 313.* |
| Describe how digoxin causes toxicity in the GI tract. | Digoxin causes toxicity in the GI tract through direct effects and by stimulation of the central nervous system, specifically vagal stimulation and chemoreceptor trigger zone stimulation.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 217.* |
| 1. Outside the heart, where does digitalis toxicity most frequently manifest? 2. List the symptoms associated with this toxicity. | 1. The GI tract. 2. Nausea, diarrhea, vomiting, anorexia.<br><br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 217.* |
| List the most common cardiac signs of digitalis toxicity. | Bigeminal rhythm, second degree AV block, AV junctional rhythm, and premature ventricular depolarizations.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 217.* |
| How does quinidine affect plasma levels of digoxin? | Quinidine increases plasma levels of digoxin in a dose-dependent manner.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 316.* |

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